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Biomass-fired power generation

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1. Introduction

Worldwide biomass ranks fourth as an energy resource, providing approximately 14% of the world's energy needs; biomass is the most important source of energy in developing nations, providing ~ 35% of their energy, particularly in rural areas where it is often the only accessible and affordable source of energy [1,2]. Considering the vastness of the photosynthetic productivity of biomass (conservatively an order of magnitude greater than the world's total energy consumption), that bioenergy can be produced and used in a clean and sustainable manner, and those technologies for biomass conversion are advancing along several fronts worldwide, there is much optimism that biomass will continue to play a significant, and probably increasing, role in the world's future energy mix. The extent of this role, especially in industrialized countries, will depend on greater exploitation of existing biomass stocks (particularly residues) and the development of dedicated energy feedstock supply systems.

2. Biomass energy potential

Biomass residues are the most readily accessible and often least costly (sometimes having a negative cost associated with their disposal) form of biomass available today. The approximate quantities of such feedstocks and their energy values are summarized, continent by continent, in Table 1 [3]. In many instances, crop processing provides much synergism with power production, the former serving to upgrade the quality of the biomass feedstock to facilitate the latter. An example is seen in sugarcane processed to produce sugar as a primary product and electricity as a by-product. Harvested sugarcane

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Table 1

Biomass residues and dedicated feedstocks for electric power generation

	Africa	North America	South America	Asia	Europe	Oceania	World
<i>Biomass residues</i>							
Selected crop residues	5[253]	15[809]	4[230]	27[1053]	10[565]	1[72]	62[3433]
Animal wastes	5[347]	3[198]	7[465]	20[1263]	4[272]	1[89]	41[2634]
Round woods	4[203]	9[429]	4[180]	14[724]	5[227]	0[23]	36[1785]
Dedicated feedstocks ^a	14[789]	10[535]	12[659]	12[668]	2[119]	5[293]	55[3036]
Total biomass energy ^b	21[1190]	23[1253]	19[1096]	43[2413]	12[651]	7[385]	12[6988]
Biomass electric potential ^c	7	8	7	15	4	2	44
Present generation, major sources	1	14	2	12	15	1	44

^aAssumes production of 5% of existing forest lands with a yield of 10 t ha⁻¹ yr⁻¹.^bConsists of dedicated feedstock plus 50% of biomass residues.^cAssumes generation efficiency = 35%.

contains > 1% alkali compounds per unit of fiber harvested. Alkalis, when reacted with sulfates and chlorine, may harm thermochemical conversion systems, fouling heat exchange surfaces, gas-turbine blades, and other power system components. The milling process that extracts sugar from the sugarcane crop produces the fibrous residue, bagasse, that is used as fuel to generate steam and electricity. The bagasse contains only 0.3% alkali per unit of fiber (< 30% the amount contained in the original sugarcane crop) as well as a lower moisture content, and the resulting fuel particle size is better suited for power generation than the original sugarcane received by the mill.

The data in Table 1 show that in some instances, particularly in developing countries, the energy contained in existing biomass residues rivals or exceeds the amount of electric energy generated from traditional sources. It must be recognized, however, that biomass residues have many applications, including use as animal feed, soil amendments, industrial feedstock, and commercial products; and that each application carries an economic value for the feedstock. Therefore, not all residues can or should be diverted to power generation. Animal wastes are another significant potential biomass resource for electricity generation, and like crop residues, have many applications, especially in developing countries.

Although residues are most widely used as sources of energy in developing nations, where such resources are often the only form of energy readily available, under favorable circumstances they can contribute significantly to the energy mix even in industrialized countries. For example, in the early 1980's, ~ 10% of the electricity generated in the state of Hawaii originated from sugarcane bagasse and in some counties of the state > 50% came from bagasse. However, if biomass is to play a major role in the world's energy mix in the longer term, crops will need to be grown specifically for energy. Studies performed by a number of investigators [4] have suggested that within 10 yr, the United States could produce large quantities (> 5 EJ) of high-yielding energy crops for US\$30–50 t⁻¹ (dry basis) or lower, potentially making such feedstocks competitive with coal in many locations. These projections will need to be validated in thousand-hectare commercial plantings. Initial installations of energy crops will need to

be profitable in order for the concept of dedicated energy feedstock supplies to gain broader acceptance.

This biomass usage potential should be considered in light of the total anticipated growth of electricity capacity additions. Between 1994–2003 capacity additions are estimated to be 629 gigawatts (GW) worldwide [5] and represents a growth of 21% over the installed base of 2925 GW now in service. Much of this growth will take place in China and Asia.

3. Status

In the United States, the period from 1973 to the present has shown a dramatic upswing in bioenergy use, especially in thermal and electrical applications of wood residues. The wood processing and pulp and paper sectors became about 70% self-sufficient in energy in this period; and the amount of grid-connected electrical capacity has increased from less than 200 MW_e in 1978 to over 7500 MW_e today. This dramatic growth, stimulated in part by Federal tax policy and state utility regulatory actions, occurred after the Public Utilities Regulatory Policies Act (PURPA) of 1978 guaranteed small electricity producers that utilities would purchase electricity at a price equal to the utilities' avoided cost.

More than 70% of biomass power is cogenerated with process heat. Wood-fired systems account for 88%, landfill gas 8%, agricultural waste 3%, and anaerobic digesters 1%. There are nearly 1000 wood-fired plants in the US, typically ranging from 10 to 25 MW_e. Only a third of these plants offer electricity for sale. The rest are owned and operated by the paper and wood products industries for their own use. Most of today's biomass grid connected power installations are the smaller scale independent power and cogeneration systems. To date, utilities have been involved in only a handful of dedicated wood-fired plants in the 40 to 50 MW_e size range, and in some cofiring of wood and municipal solid waste in conventional coal-fired plants. Net plant heat rates for 25 MW_e plants (electricity only) in the California PG&E service territory average approximately 20% efficiency (17 000 Btu kWh⁻¹). By comparison the 43 MW_e utility-operated plant at Kettle Falls, Washington, has a reported heat rate of 23.7% efficiency (14 382 Btu kWh⁻¹).

The advantageous power-purchase agreements that were negotiated under PURPA in the 1980's are no longer available at high avoided-cost rates. As a result a number of plants are closing as their power contracts come up for renewal. These plants could be competitive in today's environment using low-cost waste and residue fuels if their efficiency were much higher. This has been demonstrated in the Hawaii sugar industry where the sugar mill power plants operate for a major part of the year as combined heat and power (CHP) installations. Investments in efficient steam cycles have resulted in a competitive rate of power generation under PURPA. Low-pressure boilers were systematically replaced by higher-pressure boiler systems of larger capacity in the period 1960 through 1980, which increased the average steam pressure and temperature increasing from 1.3 MPa and 210°C to 4.4 MPa and 380°C. Meanwhile the net steam consumption in the mills decreased significantly from 600 kg tc⁻¹ (ton of cane) to about 300–400 kg

tc^{-1} , resulting in a power output of about 60 kWh tc^{-1} on average; the best mills reached over 100 kWh tc^{-1} .

4. Technology

The technologies for the primary conversion of biomass for electricity production are direct combustion, gasification, and pyrolysis. Direct combustion involves the oxidation of biomass with excess air, giving hot flue gases which are used to produce steam in the heat exchange sections of boilers. The steam is used to produce electricity in a Rankine cycle; usually, only electricity is produced in a condensing steam cycle, while electricity and steam are cogenerated in an extracting steam cycle. In air-based gasification cycles, biomass is partially oxidized by sub-stoichiometric amounts of oxygen, normally with steam present, to provide energy for thermal conversion of the remaining biomass to gases and organic vapors. For power production the cleaned gasification product gases will be fed directly to a boiler or to the combustion section of an industrial or aeroderivative turbine. In indirect gasification cycles an external heat source, instead of oxygen, is used to provide the energy for high-temperature steam gasification of the organic fraction of biomass to vapors and gases. In pyrolysis processes, indirect heating is also used to convert biomass to a mixture of gases and organic vapors. Pyrolysis is defined as the thermal destruction of organic materials in the absence of oxygen. Technically, therefore, indirect gasification is a pyrolysis process. For purposes of this discussion, if the primary pyrolysis product is gas, the process is considered gasification. If the primary products are condensable vapors, the process is considered pyrolysis. In addition, pyrolysis processes usually do not involve steam addition.

4.1. Combustion

4.1.1. Pile burner

Pile burners represent the historic industrial method [6] of wood combustion and typically consist of a two-stage combustion chamber with a separate furnace and boiler located above the secondary combustion chamber. The combustion chamber is separated into a lower pile section for primary combustion and an upper secondary combustion section. Wood is piled up to about 3.3 m (10 ft) deep on a grate in the bottom section and combustion air is fed upwards through the grate and inwards from the walls; combustion is completed in a secondary combustion zone using overfire air. Feed is introduced either on top of the pile or through an underfeed arrangement using an Auger. The underfeed arrangement gives better combustion control by introducing feed underneath the active combustion zone, but adds to system complexity and lower reliability. Ash is removed by isolating the combustion chamber from the furnace and manually dumping the ash from the grate after the ash is cooled. Pile burners typically have low efficiencies, 50 to 60%, have cyclic operating characteristics because of the ash removal, and have combustion cycles that are erratic and difficult to control. Because of the slow response time of the system and the cyclic nature of operation, pile

burners are not considered for load following operations. The advantage of the pile burner is its simplicity and ability to handle wet, dirty fuels.

4.1.2. *Stoker grate*

Stoker combustors (figure in Ref. [6]) improve on operation of the pile burners by providing a moving grate which permits continuous ash collection, thus eliminating the cyclic operation characteristic of traditional pile burners. In addition, the fuel is spread more evenly, normally by a pneumatic stoker, and in a thinner layer in the combustion zone, giving more efficient combustion. Stoker fired boilers were first introduced in the 1920's for coal, and in the late 1940's the Detroit Stoker installed the first travelling grate spreader stoker boiler for wood. In the basic stoker design the bottom of the furnace is a moving grate which is cooled by underfire air. Underfire air rate defines the maximum temperature of the grate and thus the allowable moisture content of the feed. More modern designs include the Kabliz grate, a sloping reciprocating water-cooled grate. Reciprocating grates are attractive because of simplicity and low fly ash carry-over. Combustion is completed by the use of overfire air. Furnace wall configurations include straight and bull nose water walls.

Staged combustion processes have been developed by vendors such as Zurn to meet the tighter NO_x emission limits of the 1980's. For 40% excess air the overfire air has been increased to 50% from 30%, lowering the maximum temperature in the furnace, and thus reducing NO_x levels. In the Zurn furnace the fuel on the grate burns under substoichiometric conditions at about 1150°C (2100°F), lowering the heat release by about 10% over earlier designs, and in the overfire zone volatiles burn at about 1260 to 1315°C (2300 to 2400°F). The furnace exit temperature is maintained at about 980°C (1800°F). These conditions minimize NO_x and keep furnace temperature below the ash deformation temperature of most biomass fuels. Other vendors include Foster Wheeler and Babcock and Wilcox.

4.1.3. *Bubbling fluid bed*

In a gas–solid fluidized bed a stream of gas passes upwards through a bed of free-flowing granular materials in which the gas velocity is large enough that the solid particles are widely separated and circulate freely throughout the bed. During overall circulation of the bed there will be transient streams of gas flowing upwards in channels containing few solids and clumps or masses of solids flowing downwards [7]. The fluidized bed looks like a boiling liquid and has the physical properties of a fluid. In fluidized-bed combustion of biomass the gas is air and the bed is usually sand or limestone. The air acts as the fluidizing medium and is the oxidant for biomass combustion. A fluidized-bed combustor is a vessel with dimensions such that the superficial velocity of the gas maintains the bed in a fluidized condition at the bottom of the vessel, with a change in cross-sectional area above the bed which lowers the superficial gas velocity below fluidization velocity to maintain bed inventory and act as a disengaging zone. Overfire air is normally introduced in the disengaging zone. To obtain the total desired gas-phase residence time for complete combustion and heat transfer to the boiler walls the larger cross-sectional area zone is extended and is usually referred to as the freeboard. A cyclone is used to either return fines to the bed or to

removes ash-rich fines from the system. The bed is fluidized by a gas distribution manifold or series of sparge tubes [8]. Biomass is introduced either through a feed chute to the top of the bed or through an Auger into the bed. In-bed introduction provides combustion residence time for fines which would otherwise be entrained in the fluidizing gas and not combusted in the bed. To insure complete combustion overfire air may be added to the freeboard space.

The bed is usually preheated using an external burner fired by natural gas, propane or fuel oil. The hot flue gas from the external burner is used to heat the fluidized bed to the fuel ignition temperature. For biomass this temperature is around 540°C (1000°F). Supplemental firing can be used to heat the freeboard gas to normal combustion temperature. At this point biomass is slowly introduced into the bed to raise the bed temperature to the desired operating range, normally 790 to 870°C (1450 to 1600°F). Bed temperature is governed by the desire to obtain complete combustion versus the need to maintain the bed temperature below the ash fusion temperature of the biomass ash. As biomass is introduced into the bed the majority of the organics pyrolytically vaporize and are partially combusted in the bed. The exothermic combustion provides the heat to maintain the bed at temperature and to volatilize additional biomass. Combustion is completed in the freeboard space above the bed, resulting in freeboard temperatures approaching 980°C (1800°F).

Fluidized-bed combustors have the advantage of extremely good mixing and high heat transfer, resulting in very uniform bed conditions. Combustion takes place at relatively low temperatures, giving low NO_x emissions. Combustion is very efficient, with 99–100% carbon burnout being typical [8]. Bubbling fluidized-bed boilers are normally designed for complete ash carryover, necessitating the addition of cyclones and/or bag houses for particulate control.

4.1.4. Circulating fluid bed

If the air flow of a bubbling fluid bed is increased, the air bubbles become larger, forming large voids in the bed and entraining substantial amounts of solids. This type of bed is referred to as a turbulent fluid bed [9]. In a circulating fluid bed the turbulent bed solids are collected, separated from the gas, and returned to the bed, forming a solid circulation loop. A circulating fluid bed can be differentiated from a bubbling fluid bed in that there is no distinct separation between the dense solids zone and the dilute solids zone. Circulating fluid bed densities are about 560 kg m⁻³ (35 lb ft⁻³) compared to a bubbling bed density of 720 kg m⁻³ (45 lb ft⁻³) [9]. To achieve the lower bed density air rates are increased from the 1.5–3.7 m s⁻¹ (5–12 ft s⁻¹) of bubbling beds to about 9.1 m s⁻¹ (30 ft s⁻¹) [6]. The residence time of the solids in a circulating fluid bed is determined by the solids circulation rate, the attritability of the solids, and the collection efficiency of the solids separation device. The fuel concentration in the bed is usually low, and in the case of a volatile fuel like wood the carbon content is close to zero. Even for a nonvolatile fuel like coal the carbon inventory may only be 1% [9]. As with bubbling fluid beds, the primary driving force for development of circulating fluid beds in the United States is emissions. The uniform, low combustion temperature gives low NO_x emissions. In a circulating fluid bed, with its need for introduction of solids to maintain bed inventory, it is easy to introduce a sorbent solid, such as limestone or

dolomite, to control SO_2 emissions without the need for back-end sulfur removal equipment. Circulating fluid bed temperatures are maintained at about 870°C (1600°F), which helps to optimize the limestone–sulfur reactions [10].

The major manufacturers of circulating fluid bed boilers for biomass are Combustion Engineering (CE-Lurgi), B&W-Studsvik, Ahlstrom Pyropower (Foster Wheeler), and Gotaverken. A number of plants have been built in the 25 MW size range, primarily in California. A major advantage of circulating fluid bed boilers is the ability to handle varying feedstocks with different compositions and moisture contents. As with bubbling-bed boilers, bed agglomeration is a concern. High alkaline content fuels cause particles in the bed to agglomerate and defluidize, eventually plugging the system.

4.1.5. Suspension burners

The suspension burning of pulverized wood in dedicated biomass boilers is a fairly recent development and is practised in relatively few installations. Suspension burning has also been accomplished in lime kilns [11] and is being investigated by the utility industry for cofiring applications [12]. The requirements for successful suspension firing, a feed moisture content of less than 15% [6] and a particle size less than 0.0015 m (0.06 in.) [11] give higher boiler efficiencies, up to 80%, than firing wet wood chips, 50–55% moisture, in a stoker grate or fluid bed, at 65% efficiency. The higher efficiency also results in smaller furnace size. Offsetting the higher efficiency is the cost and power consumption of drying and comminution. In addition, special burners need to be used. Burners developed for suspension firing include scroll cyclonic burners and vertical-cylindrical burners [6].

Installations include the Oxford Energy, 27 MW, facility at Williams California [6], the ASSI Lvholmen Linerboard Mill in PiteD [13], the Klabin do Parana mill in Monte Alegre, Brazil [11], and the E.B. Eddy Mill in Espanola Ontario [11].

4.1.6. Whole Tree Energy™

The Whole Tree Energy™ Process is being developed by Energy Performance Systems, Minneapolis, MN [14], as an integrated wood conversion process encompassing feedstock production, harvesting, transportation, and conversion to electricity. The concept involves burning whole trees in a staged combustion system with the use of low temperature gas to dry the trees. Elements of the process have been tested, but the system has not been run as an integrated process.

The concept involves transporting whole trees to the conversion facility where drying will be accomplished over a 30-day period using low temperature heat from the power island. For a 100-MW facility the drying building is envisioned as a dome approximately 61 m (200 ft) high and 198 m (650 ft) in diameter [15]. Trees will be stacked in a circular doughnut fashion around a central tower crane which is used to unload trucks and to deliver feed to a drag conveyor for transport to the boiler. Trees will be stacked to about a 12-m (40 ft) height. The drying will be accomplished using a hot air ducting system located underneath the wood pile.

Trees will be transported to the power island where they will be cut to the desired length, 8.5–9.1 m (28–30 ft), and introduced into the combustion chamber through a ram charger door into the primary combustion chamber. The primary combustion

chamber is envisioned as a deep bed about 8.5 m (28 ft) tall with whole trees supported on a water-cooled grate. This bed will be operated as a substoichiometric combustor to produce a mixture of combustion products and volatilized organics. The gases leaving the primary combustion chamber will be burned with overfire air under excess air conditions to complete the combustion process. Char falling through the grate in the deep bed will also be burned to minimize organics in the ash. The excess air rate is estimated at 13%, and the overfire to underfire air ratio will be about 30/70.

The boiler will be a standard design with superheater and economizer. The flue gases leaving the economizer will pass through a plate type condensing heat exchanger to preheat combustion air to 370°C (700°F) and drying air to 54°C (130°F). The steam turbine cycle will be comparable to modern cycles utilizing 16.54 MPa (2400 psia), 538°C (1000°F) steam.

The potential advantages for the Whole Tree Energy™ process are reduced operating costs achieved by elimination of wood chipping, and increased efficiency by almost complete use of waste heat in the condensing heat exchange system. The net heat rate for a 100 MW facility is projected to be about 11 236 KJ kWh⁻¹ (10 650 Btu kWh⁻¹).

4.1.7. Cofiring with coal

A number of major utilities are evaluating the cofiring of biomass in existing coal-fired power stations because of the perceived environmental benefits which may accrue [12]. These benefits include the reduction in carbon dioxide emissions from the combustion of fossil fuels, the reduction of SO₂ formation through a decrease in fuel bound sulfur, the reduction of NO_x formation through a reduction in fuel bound nitrogen, and a means to address air toxics emissions should the Clean Air Act Amendments Title III exemption for utilities be eliminated. Incentives which exist for utilities include site specific cases where biomass purchase price is less than coal; indirect benefits to the forestry and secondary wood products industry by providing new markets for wastes and residues, as well as dedicated feedstocks, thereby promoting economic stability in the utilities service area; and the potential to develop feedstock infrastructure and jobs for long term biomass development and use.

A number of utilities have instituted commercial cofiring operations or have performed commercial scale cofiring tests in utility boilers. These operations have covered the majority of types of utility boilers that are expected to be candidates for cofiring and include cyclone units, pulverized coal units (suspension firing) and fluidized bed units. The utilities include Northern States Power (NSP), Delmarva, Georgia Power, Santee Cooper, Savannah Electric, and the Tennessee Valley Authority (TVA). For example, NSP in Minnesota has investigated cofiring biomass in a number of their coal-fired boilers, including King station (a cyclone unit), French Island (a fluidized bed unit), Black Dog (a fluidized bed unit), and Granite Falls (a pulverized coal unit). NSP routinely co-fires 200 000–300 000 t yr⁻¹ of biomass. TVA has recently completed cofiring tests at their Allen Station, in a 250 MW cyclone unit, and at the Kingston station, in a 200 MW tangentially-fired pulverized coal unit. The tests at the Allen station were performed at a 10% by mass level and demonstrated the potential for significant technical, economic, and environmental benefits associated with wood cofiring, including improving dust control during fuel handling, achieving 100% boiler

capacities while cofiring, experiencing minimum impacts on boiler efficiency, and achieving reductions in pollutants such as NO_x . Low level cofiring tests were performed at the Kingston station by cofeeding wood with coal to a pulverizer at a 4–5 mass% level and found that at this level acceptable pulverizer power consumption and performance could be achieved.

Based on utility experience with cofiring some general conclusions [12] have been reached for application to utility boilers. For low levels of cofiring, up to 5–8% biomass, wood can be combined with coal prior to the pulverizers. At medium levels, 10 to 15% biomass, a separate wood preparation and delivery system should be provided, with boiler operation based on coal characteristics. At moderately high levels, 25 to 50%, a multifuel system such as a fluid bed should be used. At high levels, above 50%, a boiler designed specifically for biomass operation should be used.

4.2. Gasification

Gasification for power production involves the devolatilization and conversion of biomass in an atmosphere of steam and/or air to produce a medium or low calorific value gas. If air is present, the ratio of oxygen to biomass is typically around 0.3. A large number of variables affect gasification based process design. Gasification medium is an important variable. Air blown, or directly heated gasifiers, use the exothermic reaction between oxygen and organics to provide the heat necessary to devolatilize biomass and to convert residual carbon-rich chars. In these directly heated gasifiers, the heat to drive the process is generated internally within the gasifier. Thus, when air is used, the resulting product gas is diluted with nitrogen and typically has a dry basis calorific value of about $4.9\text{--}5.9 \text{ MJ Nm}^{-3}$ at 273 K ($125\text{--}150 \text{ Btu scf}^{-1}$ at 530 R). The dry basis calorific value of the product gas can be increased to $12.8\text{--}13.8 \text{ MJ Nm}^{-3}$ ($300\text{--}350 \text{ Btu scf}^{-1}$) through the use of oxygen instead of air. Oxygen production is expensive however, and its use has only been proposed for direct heating gasification applications involving the production of synthesis gas where nitrogen is not permitted in downstream synthesis conversion operations. Oxygen typically costs US\$40 to US\$60 Mg^{-1} (US\$36–54 t^{-1}), and typically is used at the rate of 0.25 to 0.3 Mg Mg^{-1} of biomass, thus translating to a cost equivalent to US\$10 to US\$20 Mg^{-1} (US\$9–18 t^{-1}) [16] of biomass. Indirectly heated gasifiers accomplish biomass heating and gasification through heat transfer from a hot solid or through a heat transfer surface. Since air is not introduced into the gasifier, no nitrogen diluent is present and a medium calorific gas is produced, with dry basis values of $17.7\text{--}19.7 \text{ MJ Nm}^{-3}$ ($450\text{--}500 \text{ Btu scf}^{-1}$) being typical.

A second variable affecting gasification-based power systems performance is gasifier operating pressure. The majority of gasification power applications will involve a gas turbine-based generation cycle. Turbines typically operate at compression ratios of 10 to 20, giving turbine inlet pressures of 1.0 to 2.5 MPa (150 to 375 psia). A pressurized gasifier will produce gas at a pressure suitable for direct turbine application and provide the highest overall process efficiency. To take full advantage of operating a pressure, however, a number of ancillary systems must be developed. Reliable, high pressure feed systems have not been commercially proven. Secondly, at typical gasifier conditions,

825°C (1517°F) and 2.07 MPa (300 psia), tars, chars and volatile alkalis are generated. To maximize system efficiency, these materials must be removed from the hot product gas without lowering the temperature below the tar dew point, typically about 538°C (1000°F). Thus, hot-gas cleanup systems are required. While tars have relatively high heat contents and can be burned in combustors, they may cause plugging of char filters and may form soot during combustion, thus affecting combustion stability. Therefore, the first element of a hot-gas cleanup system will probably be a catalytic or thermal tar cracker. A catalytic tar cracker will operate at temperatures comparable to gasifier temperature, about 825°C (1517°F), while a thermal cracker will typically operate in a temperature range of 871–982°C (1600–1800°F). After the tar cracker the product gas will be partially cooled to minimize the amount of alkali vapors, typically to 538–649°C (1000–1200°F). The product will then pass through a ceramic filter to remove solids. Much of the alkali will also be removed. For gas turbine applications the gas may still contain too much alkali. Gas turbine limits for alkali are about 20 ppb in the turbine combustor exit gas. To insure the product gas meets gas turbine manufacturer alkali specifications, an alkali getter bed may be added after the filter. Typical bed materials are emalthisite or hectorite. Getter beds will probably be designed as parallel fixed beds, giving continuous operation while allowing for bed material replacement.

Alternatively, the gasifier can be operated at low pressure and the cleaned product gas compressed to the pressure required for gas turbine application. In this case a tar cracker will probably be used to minimize the amount of tars which must be handled during quenching. The product gas exiting the tar cracker will be conditioned to provide a suitable compressor feed. Gas characteristics to be considered for compressor use include the amount of condensible or soluble tars and the water vapor content. The water and tar content must be low enough to insure no condensation during compression. Also, soluble tars which will impact lubrication oil properties must be eliminated. Lastly, the gas inlet temperature must be suitable for compressor materials of construction. Normally, the temperature should be held below about 93°C (200°F). Usually a combination of heat exchange, to reduce the gas from tar cracker exit temperature to residual tar dewpoint, and wet scrubbing will be used. The water vapor content will be at saturation at scrubber exit temperature and pressure.

The other major variable is reactor type. For biomass gasification four primary types of reactor systems have been developed: fixed bed reactors, bubbling fluid–bed reactors circulating fluid–bed reactors, and entrained flow reactors. Gasification reactors operate under much of the same principles as comparable combustors.

4.2.1. IGT/Tampella

The Institute of Gas Technology (IGT) has developed the RENGAS gasification technology specifically for the conversion of biomass to low- or medium-heating value gas [17]. Biomass is fed to a single pressurized, bubbling, fluidized bed-gasifier vessel. Inert alumina beads form the deep fluidized bed and provide stable fluidization behavior and needed heat capacity for efficient transfer of energy released by the combustion to endothermic devolatilization and gasification reactions. The use of a deep single-stage bed of inert solids yields high carbon conversion. Feed is introduced into the fluidized bed by means of a high speed screw. The process has been tested during more than 250

h of steady-state operation at up to a 10.9 Mg day^{-1} (12 tpd) feed rate and at pressures up to 3.45 MPa (500 psia) in a 0.292 m (0.96 ft) diameter by 3.1 m (10.2 ft) high fluidization zone. The disengaging zone dimensions are 0.457 m (1.5 ft) diameter by 3.35 m (11 ft) in height. Parameters studied have included gasification temperature, pressure, feed moisture, feedstock type, steam input, bed media, fluidized bed height and gas superficial velocity. Biomass feedstocks gasified include maple wood chips, whole tree chips, California highway clippings, paddy rice straw, refuse-derived fuel, bark and paper mill sludge, bagasse from Hawaii, and alfalfa. The RENGAS technology is being scaled up to 91 Mg day^{-1} (100 tpd) by the Pacific International Center for High Technology Research at the Hawaii Commercial and Sugar Paia, Maui, Hawaii, sugar mill, where the gasifier will be operated in conjunction with a Westinghouse Electric ceramic filter system. Phase 1 of the Hawaii project has been completed, with the gasifier (without hot-gas cleanup) being operated for over 100 h at throughputs up to 50 tpd and pressures up to 150 psig. Tampella Power has operated their variation of the IGT technology for a number of years at their pilot facility in Tampere, Finland at up to 80 tpd.

4.2.2. Ahlstrom (Foster Wheeler)

A. Ahlstrom OY, Finland (now owned by Foster Wheeler) has been developing a high-pressure circulating-fluid-bed gasifier for a number of years. This development has culminated in the construction and operation of a demonstration unit in Varnamo, Sweden. Not much public information is known about the operation of the unit. The capacity of the unit is approximately 90 tpd and uses ceramic filters for hot-gas clean up. In combined cycle operation the generating capacity of the plant is about 4 MW from a European Gas Turbine Typhoon series gas turbine and 2 MW from a bottoming steam cycle. Approximately 9 MW thermal of hot water is also produced for district heating. The gasifier started operating in 1993, and in the fall of 1995 the first turbine operations commenced.

4.2.3. Thermiska processor TPS

The TPS gasifier process is a low pressure, air-blown, circulating fluid bed design in which the primary gasification step is followed by second circulating fluid bed cracker. In the cracker dolomite (mixed magnesium–calcium carbonate) is used to catalyze the conversion of tars to gases and lower molecular weight vapors. Tar cracking reduces the amount of organics removed from the product gas stream during water quenching operations, and thus maximizes useful product recovery and minimizes water treatment costs. The TPS process has been in development for a number of years. In 1992, Ansaldo Aerimpianti of Italy installed a commercial, two-bed unit in Greve-in-Chianti, Italy. The Greve units have a combined capacity of 30 MWt burning pelletized refuse derived fuel. Limited operations have also been performed using hogged wood or agricultural wastes. Fuel gas generated at the plant is either burned in a boiler to produce steam for electricity production or used as fuel in a nearby lime kiln operation.

TPS has been selected to supply technology for a biofuel based combined cycle power plant in Brazil. The project, cofunded by the World Bank Global Environment Facility and a number of Brazilian utilities, lead by Electrobras, will produce electricity

from eucalyptus in a 30-MW combined cycle facility. TPS technology is also being used for a 5-MW demonstration project in Yorkshire in the United Kingdom.

Similar low-pressure, air-blown gasifiers have also been developed by Lurgi [18] and Ahlstrom.

4.2.4. Battelle Memorial Laboratory

Battelle Memorial Laboratory [19,20], Columbus, OH, has been developing an indirectly heated biomass gasification process, called the Battelle High Throughput Gasification Process, since 1977. A 10 tpd gasifier has been operating in West Jefferson since 1980. The Battelle gasification process produces a medium-Btu gas without using air or oxygen in the gasification step. The process uses two physically separate reactors, a gasification reactor in which biomass is converted into a medium Btu gas and residual char, and combustion reactor that burns the residual char to heat sand which circulated back to the gasifier to provide heat for gasification. The Battelle process utilizes circulating fluid bed reactors to take advantage of the inherently high reactivity of biomass feedstocks and has demonstrated throughputs in excess of $3000 \text{ lb h}^{-1} \text{ ft}^{-2}$.

The Battelle process is being scaled to 200 tpd (dry basis) as part of a major USDOE initiative to demonstrate gasification of renewable biomass for electricity production. The demonstration and validation of this gasification/gas turbine system is being undertaken at the existing 50 MW wood-fired McNeil Power Generating Station in Burlington, Vermont. The first phase of the project involves design, construction and operation of the gasifier on wood chips. This phase is underway, with commissioning of the gasifier expected in the spring of 1997. Installation of a 15-MW gas turbine system is anticipated in late 1997 or early 1998.

5. Economics

In policy analysis literature, all of the economic analyses are performed in terms of the N th plant—an ideal plant that represents the ultimate economic performance for that technology. Over the years, a large number of organizations have projected this ultimate performance of biomass-based power systems. These projections, however, often covered a broad range which lead inevitably to reluctance on the part of developers to invest in these technologies. In fact, using today's technology for plant 1, the maximum scale that would be risked is probably about 18 MW_e , efficiency in the range 30–33%, with a capital investment of $> \text{US\$}2500 \text{ kW}^{-1}$.

Recently, the US Environmental Protection Agency (EPA) embarked on a study to evaluate the penetration of various technologies into the power generation marketplace in the coming decades and their effect on carbon emissions to the atmosphere. This study [21] required as input cost and efficiency estimates for biomass, coal, and natural gas power generation systems. Given the disparity of opinion on these figures in the published literature, a panel consisting of representatives from NREL, the Electric Power Research Institute (EPRI), the Princeton Center for Energy and Environmental Studies, EPA, the United States Department of Agriculture (USDA), and the Colorado School of Mines was convened to arrive at a consensus position. The panel was able to

Table 2
Technology characterizations used for EPA modeling

Component	Industrial turbine		Advanced turbine systems		
	Biomass	Coal	Biomass	Coal	Natural gas
<i>'Low' technology specifications</i>					
Heat rate (Btu kWh ⁻¹)	8660	8700	7579	7614	6202
Efficiency (% HHV)	39.4	39.2	45.0	44.8	55.0
Fixed O + M (US\$ kW ⁻¹)	51.25	51.25	39.66	39.66	28.80
Variable O + M (mils kWh ⁻¹)	3.15	3.15	2.46	2.46	0.712
Total capital (US\$ kW ⁻¹)	1230	1254	1023	1047	522.50
<i>'High' technology specifications</i>					
Heat rate (Btu kWh ⁻¹)	9400	8700	8227	7614	6202
Efficiency (% HHV)	36.3	39.2	41.5	44.8	55.0
Fixed O + M (US\$ kW ⁻¹)	44.71	36.44	34.60	28.20	28.80
Variable O + M (mils kWh ⁻¹)	3.65	2.60	2.85	2.03	0.712
Total capital (US\$ kW ⁻¹)	1488	1254	1243	1047	522.50

agree upon likely ranges of cost and performance for system utilizing these fuels. Table 2 summarizes these ranges.

Fig. 1 correlates [22] a number of studies and cost estimate efforts. The lines on this figure represent 0.8 scale factor curves using, as a base point, the high and low technology estimates from the EPA study. Plotted as points around the curves are the cost estimates from the various studies shown. The GEF study is the Global Environ-

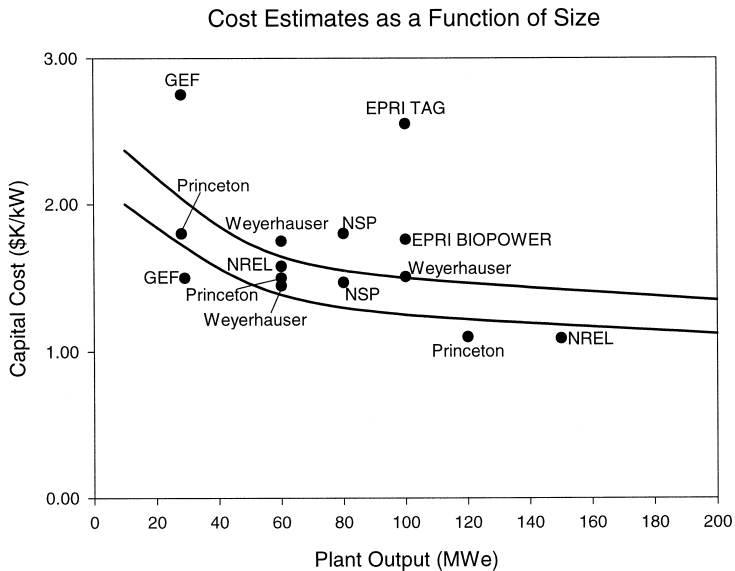


Fig. 1. Capital cost estimates as a function of plant size.

Table 3
New small cogeneration plants in Finland

Plant	Takeover	Capacity power/heat (MW _e MWt ⁻¹)	Steam conditions (EC bar ⁻¹)	Type of boiler	Investment cost (Million ECU)
Pieks-m-ki	1992	9/25	510/60	BFBC	9.5
Kankaanp	1992	6/16	510/60	BFBC	8.0
Kuhmo	1992	5/13	490/81	CFBC	12.3
Ylivieska	1993	5/15	510/60	BFBC	8.7
Kuusamo	1993	6/17	510/60	BFBC	8.0
Lieksa	1994	8/22	510/61	CFBC	11.3

ment Fund facility in Brazil that is using TPS Thermiska Processor gasification technology and a General Electric aero-derivative turbine [23]. The other studies are listed in the reference section [24–27]. These studies demonstrate the likelihood that biomass system cost and performance can achieve levels that make them competitive or otherwise attractive in existing niche markets.

A recent paper [28] has presented investment costs for small cogeneration plants in Finland. These costs (Table 3) are for bubbling fluid bed and circulating fluid bed combustor systems less than 10 MW_e producing electricity and district heat and/or industrial process heat. These systems are primarily fueled by peat and wood wastes and power-to-heat ratios of about 0.35.

6. Environmental benefits

Significant environmental benefits can be obtained by using biomass fuels in direct combustion, gasification, or pyrolysis systems, although some uncertainties exist at present. SO₂, CO₂, and ash production will be typically be far lower for biomass power systems than for coal combustion and conversion systems. Biomass power is one of the most attractive options for addressing CO₂ concerns, because both growth and conversion involve recycling atmospheric carbon, resulting in no net addition of CO₂ into the atmosphere. Emissions of air toxics from biomass direct or indirect combustion requires further characterization, but they will probably be less of a problem than the air toxics emissions from coal combustion. Water quality impacts should be smaller for biomass-fueled systems than for coal-fueled systems, even though water usage in biomass combustion is comparable to that used in coal combustion. However, feedstock growth could require significant use of water and petrochemical-based fertilizer, and that raises concern about nutrient run-off. Solid waste, in the form of ash, is generally viewed as non-hazardous and is produced in smaller quantity than in coal-fueled systems. Solid waste production is lower because of the relatively low ash content of biomass fuel. Long-term ecological effects such as destruction of wildlife habitat, loss of biodiversity, and sustainability of soil productivity are other issues unique to the biomass fuel resource and deserve further attention.

7. Fields of application

Biomass-based power systems are unique among nonhydro renewable power sources because of their wide range of applicability to a diverse set of needs. Biomass systems can be used for village-power applications in the 10–250 kW scale, for larger scale municipal electricity and heating applications, for industrial application such as hog-fuel boilers and black-liquor recovery boilers, in agricultural applications such as electricity and steam generation in the sugar cane industry, and for utility-scale electricity generation in the 100 MW_e scale. Biomass-based systems are the only nonhydro renewable source of electricity that can be used for base-load electricity generation.

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